HANDS-ON LAB 01 GUIDE & COPY-PASTE PROMPTS • TELECOMMUNICATIONS EDITION (TELECOM NEXUS)
Delivered by Tryolabs • Powered by Google Antigravity & Gemini Enterprise

TRYOLABS • GOOGLE ANTIGRAVITY WORKSHOP • LAB 01 (EN)

Lab 01: 5G Slice SLA & CDR Revenue Assurance Triage

Antigravity Capabilities Practiced: @ Context Mentions, **`/plan`** Mode, Inline Command (**`Cmd+I`** / **`Ctrl+I`**), Antigravity Tab (Supercomplete), Visual Diff Overlays, Terminal Test Execution, **`/browser`**, and **`/learn`**.

Language / Idioma: English | Español • **Prompt Playbook:** PROMPTS.md | PROMPTS.es.md • **PDF:** EN (Dark) | EN (Light) | ES (Dark) | ES (Light)

1. Scenario & Business Context

You are a Senior Principal Telecom OSS/BSS Engineer on call for **Telecom Nexus**. Two critical alarms just fired in **`data/logs/ran_alarms.jsonl`**:

1. **`ALM-BSS-8006` (Revenue Assurance Anomaly)**: Enterprise B2B data sessions are being severely underbilled.

- In src/services/cdr_rating.py, **`BYTES_DIVISOR`** was mistakenly set to **`Decimal(1024 ** 4)`** (Terabytes!) instead of **`Decimal(1024 ** 3)`** (Gigabytes / GiB per **3GPP TS 32.298**).
- Additionally, visiting PLMN roaming markup (**`roaming_plmn != "310-410"`**) is only applied when **`rat_type == "EUTRAN_4G"`**, silently dropping roaming charges on all **5G Standalone (`NR_5G`)** roaming CDRs!

2. **`ALM-3GPP-8001` .. `ALM-3GPP-8005` (5G Network Slice SLA Breaches)**:

- In src/services/sla_monitor.py, **`evaluate_contract_sla_compliance()`** fails to filter **`WHERE status = 'ACTIVE'`** (accidentally penalizing **`TERMINATED`** contract **`CTR-B2B-1011`** and **`DRAFT`** contract **`CTR-B2B-1012`**).
- Worse, the latency breach condition is inverted (**`worst_latency < committed_latency`** instead of **`worst_latency > committed_latency`**), marking healthy low-latency slices as breached while ignoring congested cells!

2. Lab Setup

Run the Lab 01 installer from the repository root to ensure a clean starting state:

```
./labs/lab-01-revenue-sla-triage/install.sh
```

Confirm that the starter bug causes the Lab 01 test suite to fail:

```
./.venv/bin/pytest tests/test_lab01_triage.py
```

Optional — launch the Live Telecom Nexus Web Dashboard in a second terminal:

```
./.venv/bin/uvicorn src.main:app --reload --port 8001
```

Open **`http://127.0.0.1:8001`** (or **`http://127.0.0.1:8001/es`** for Spanish, or use the **`EN | ES`** toggle in the top-right header) to inspect the live telemetry and SLA table.

3. Step-by-Step Instructions with Google Antigravity

Step 3.1: Diagnose Root Causes with @ Mentions

Open Antigravity and paste the **Discovery Prompt** from PROMPTS.md:

```
@data/logs/ran_alarms.jsonl @src/services/cdr_rating.py @src/services/sla_monitor.py @te \
  sts/test_lab01_triage.py
We have two critical OSS/BSS incidents reported in our 3GPP alarm stream (`ALM-BSS-8006` \
   and `ALM-3GPP-8001`..`8005`):
1. Our GSMA TAP3 / 3GPP CDR rating engine is underbilling enterprise data sessions and m \
  issing 5G Standalone (`NR_5G`) roaming markups.
2. Our B2B 5G Network Slice SLA monitor is misreporting breached contracts and including \
   non-active contracts.

Run `./.venv/bin/pytest tests/test_lab01_triage.py` to reproduce the failures, then expl \
  ain the exact root causes in `cdr_rating.py` and `sla_monitor.py`.
```

Step 3.2: Generate & Review a Plan with `/plan`

Ask Antigravity to plan the fix before editing files:

```
/plan Create a step-by-step remediation plan to fix `src/services/cdr_rating.py` and `sr \
  c/services/sla_monitor.py` so that all tests in `tests/test_lab01_triage.py` pass.
```

Step 3.3: Fix `src/services/cdr_rating.py` and `src/services/sla_monitor.py`

Either approve the **`/plan`** in Chat Canvas or use **`Cmd+I`** / **`Ctrl+I`** inline in the editor:

1. In **`src/services/cdr_rating.py`**:

- Change **`BYTES_DIVISOR = Decimal(1024 ** 4)`** to **`BYTES_PER_GB = Decimal(1024 ** 3)`**.
- Change **`if roaming_plmn != HOME_PLMN and rat_type == "EUTRAN_4G":`** to **`if roaming_plmn != HOME_PLMN:`**.

2. In **`src/services/sla_monitor.py`**:

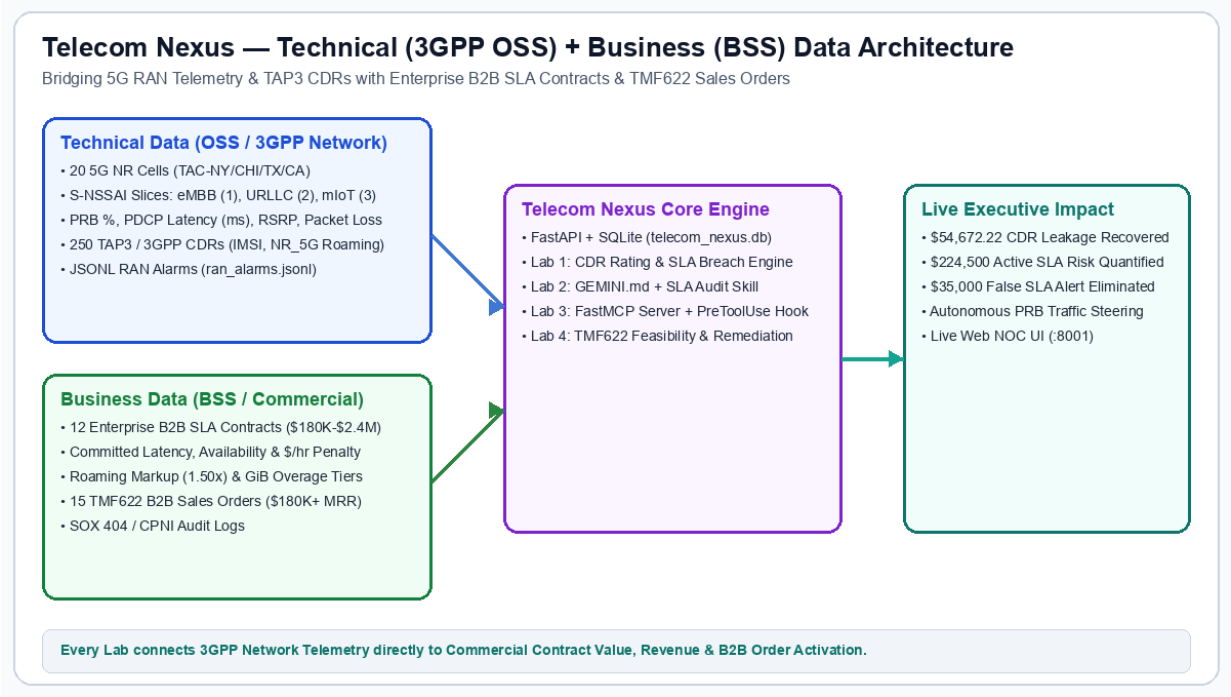
- Filter contracts by **`WHERE status = 'ACTIVE'`**.
- Change **`latency_breach = worst_latency < committed_latency`** to **`latency_breach = worst_latency > committed_latency`**.

4. Verification

Run the Lab 01 verification script:

```
./labs/lab-01-revenue-sla-triage/verify.sh
```

Need the Reference Solution? Run **`./labs/lab-01-revenue-sla-triage/apply_solution.sh`** at any time.


Telecom Nexus — Technical (3GPP OSS) + Business (BSS) Data Architecture
Bridging 5G RAN Telemetry & TAP3 CDRs with Enterprise B2B SLA Contracts & TMF622 Sales Orders
Technical Data (OSS / 3GPP Network)
• 20 5G NR Cells (TAC-NY/CHI/TX/CA)
• S-NSSAI Slices: eMBB (1), URLLC (2), mIoT (3)
• PRB %, PDCP Latency (ms), RSRP, Packet Loss
• 250 TAP3 / 3GPP CDRs (IMSI, NR_5G Roaming)
• JSONL RAN Alarms (ran_alarms.jsonl)
Business Data (BSS / Commercial)
• 12 Enterprise B2B SLA Contracts ($180K-$2.4M)
• Committed Latency, Availability & $/hr Penalty
• Roaming Markup (1.50x) & GiB Overage Tiers
• 15 TMF622 B2B Sales Orders ($180K+ MRR)
• SOX 404 / CPNI Audit Logs
Telecom Nexus Core Engine
• FastAPI + SQLite (telecom_nexus.db)
• Lab 1: CDR Rating & SLA Breach Engine
• Lab 2: GEMINI.md + SLA Audit Skill
• Lab 3: FastMCP Server + PreToolUse Hook
• Lab 4: TMF622 Feasibility & Remediation
Live Executive Impact
• $54,672.22 CDR Leakage Recovered
• $224,500 Active SLA Risk Quantified
• $35,000 False SLA Alert Eliminated
• Autonomous PRB Traffic Steering
• Live Web NOC UI (:8001)
Every Lab connects 3GPP Network Telemetry directly to Commercial Contract Value, Revenue & B2B Order Activation.

LAB 01 COPY-PASTE ANTIGRAVITY PROMPTS (EN)
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Lab 01 — Optimized Antigravity Prompts Cheat Sheet (English)

Language / Idioma: English | Español • **Lab Guide:** README.md | README.es.md • **PDF:** EN | ES

Goal: Use **Google Antigravity** (Antigravity 2.0 Desktop App, Antigravity IDE, or **`agy`** CLI) to diagnose and repair two critical production incidents in **Telecom Nexus**:

1. **CDR Revenue Leakage ($54,000+ unbilled)** in **`src/services/cdr_rating.py`** (3GPP TS 32.298 / GSMA TAP3.12).

2. **Inverted 5G Slice SLA Latency & Contract Status Filter Bug** in **`src/services/sla_monitor.py`** (3GPP TS 28.552).

1. Discovery & Root-Cause Triage Prompt (Use @ Mentions)

Copy and paste this into the **Antigravity Chat Canvas / Sidebar Chat**:

```
@data/logs/ran_alarms.jsonl @src/services/cdr_rating.py @src/services/sla_monitor.py @te \
  sts/test_lab01_triage.py
We have two critical OSS/BSS incidents reported in our 3GPP alarm stream (`ALM-BSS-8006` \
   and `ALM-3GPP-8001`..`8005`):
1. Our GSMA TAP3 / 3GPP CDR rating engine is underbilling enterprise data sessions and m \
  issing 5G Standalone (`NR_5G`) roaming markups.
2. Our B2B 5G Network Slice SLA monitor is misreporting breached contracts and including \
   non-active contracts.

Run `./.venv/bin/pytest tests/test_lab01_triage.py` to reproduce the failures, then expl \
  ain the exact root causes in `cdr_rating.py` and `sla_monitor.py`.
```

2. Planning Mode Prompt (`/plan`)

Use **`/plan`** to review Antigravity's step-by-step remediation strategy before any code is modified:

```
/plan Create a step-by-step remediation plan to fix `src/services/cdr_rating.py` and `sr \
  c/services/sla_monitor.py` so that:
1. `cdr_rating.py` converts `uplink_bytes + downlink_bytes` to binary Gigabytes (`Decima \
  l(1024 ** 3)`) instead of Terabytes (`1024 ** 4`) and applies `roaming_markup_pct` whe \
  never `roaming_plmn != "310-410"` across all `rat_type` values (`NR_5G` and `EUTRAN_4G \
  `), rounding with `Decimal` `ROUND_HALF_UP` to `0.01`.
2. `sla_monitor.py` queries only `status = 'ACTIVE'` contracts from `b2b_contracts` and  \
  flags `latency_breach = worst_latency > committed_latency`.
3. Verify the fix by running `./labs/lab-01-revenue-sla-triage/verify.sh`.
```

3. Instructive Inline Edit Prompts (`Cmd+I` / `Ctrl+I` in Antigravity IDE)

If you want to fix each bug using **Inline Command** directly inside the editor:

A. Inside `src/services/cdr_rating.py` (Highlight lines 19-48 and press `Cmd+I` / `Ctrl+I`)

```
Fix BYTES_DIVISOR to use binary Gigabytes `Decimal(1024 ** 3)` per 3GPP TS 32.298, and a \
  pply the roaming markup multiplier whenever `roaming_plmn != HOME_PLMN` regardless of  \
  `rat_type` (including `NR_5G`). Keep all arithmetic in `Decimal` rounded via `round_mo \
  ney()`.
```

B. Inside `src/services/sla_monitor.py` (Highlight lines 29-65 and press `Cmd+I` / `Ctrl+I`)

```
Filter `b2b_contracts` so only rows with `status = 'ACTIVE'` are queried, and fix `laten \
  cy_breach` so a breach occurs when `worst_latency > committed_latency`.
```

4. Full Autonomous Implementation & Verification Prompt

```
Apply the fixes to @src/services/cdr_rating.py and @src/services/sla_monitor.py, then ru \
  n `./labs/lab-01-revenue-sla-triage/verify.sh` and print a summary table showing:
1. Total CDRs audited, underbilled CDR count, and total recovered revenue leakage ($ USD \
  ).
2. The 5 breached ACTIVE B2B contracts, their degraded 5G RAN cell IDs, observed PDCP la \
  tency vs committed SLA latency, and exact SLA penalty exposure ($224,500.00 total).
```

5. Power-User & Stretch Prompts (`/grill-me`, `/browser`, `/learn`)

Interactive Design Interview (`/grill-me`)

```
/grill-me Interview me on how we should evolve our 5G Standalone CHF rating engine to su \
  pport tiered overage buckets, mid-session QoS 5QI downgrades, and bilateral GSMA BCE ( \
  Billing and Charging Evolution) settlement with visiting PLMNs 234-15 and 262-01.
```

Live NOC Dashboard Inspection (`/browser`)

```
/browser Start the FastAPI server (`./.venv/bin/uvicorn src.main:app --port 8001`), open \
   `http://localhost:8001`, and verify that the KPI cards display the recovered CDR Reve \
  nue Leakage and $224,500.00 Active B2B SLA Penalty Exposure.
```

Persist Domain Knowledge (`/learn`)

```
/learn Remember that in Telecom Nexus, byte-to-GiB conversion must always use `Decimal(1 \
  024 ** 3)`, roaming markup applies to all visiting PLMNs (`!= '310-410'`) across both  \
  `NR_5G` and `EUTRAN_4G`, and SLA penalty exposure only applies to `ACTIVE` B2B contrac \
  ts.
```